

• Our basis vectors are

$$\hat{e}_x = (1, 0, 0)$$

$$\hat{e}_y = (0, 1, 0)$$

$$\hat{e}_z = (0, 0, 1)$$

• So, the solutions are just being very pedantic:

Recall for a general vector

$$\vec{A} = (A_x, A_y, A_z)$$

Thus, the notation from the solutions are

$$\begin{aligned}\hat{e}_x &= ((\hat{e}_x)_x, (\hat{e}_x)_y, (\hat{e}_x)_z) \\ &= (1, 0, 0)\end{aligned}$$

so

$$1 = (\hat{e}_x)_x$$

$$0 = (\hat{e}_x)_y$$

$$0 = (\hat{e}_x)_z$$

How to 'read' this:

$(\hat{e}_x)_z$: This is the z -component of the basis x -vector $\hat{e}_x$

and likewise for all others:

If you're more comfortable with the other notation, this is

$$\vec{A} = (A_x, A_y, A_z) = A_x \hat{e}_x + A_y \hat{e}_y + A_z \hat{e}_z$$

so for

$$\hat{e}_z = (0, 0, 1) = \underset{\uparrow}{0} \hat{e}_x + \underset{\uparrow}{0} \hat{e}_y + \underset{\uparrow}{1} \hat{e}_z$$

$$0 = (\hat{e}_z)_x, \quad 0 = (\hat{e}_z)_y, \quad 1 = (\hat{e}_z)_z$$

$$\hat{e}_y = (0, 1, 0) = 0\hat{e}_x + 1\hat{e}_y + 0\hat{e}_z$$

$$(\hat{e}_y)_x = 0$$

$$(\hat{e}_y)_y = 1$$

$$(\hat{e}_y)_z = 0$$

$$* r(\varphi) = a e^{k\varphi}$$

$$\varphi = \omega t$$

$$r(\omega t) = a e^{k\omega t}$$

$$z = r(\omega t) e^{i\omega t} = a e^{i\omega t + k\omega t} \\ = a e^{\omega t(i+k)}$$

$$* \frac{dz}{dt} = a \omega (i+k) e^{\omega t(i+k)} = \omega (i+k) z$$

$$\frac{d^2 z}{dt^2} = a \omega^2 (i+k)^2 e^{\omega t(i+k)}$$

$$= \omega (i+k) \frac{dz}{dt}$$

$$\frac{d^2 z}{dt^2} = \frac{dz}{dt} \frac{1}{z} \frac{dz}{dt}$$

$$\circ \frac{d^2 z}{dt^2} \approx \frac{1}{z} \left(\frac{dz}{dt} \right)^2 = F(z)/m$$

$$F = -\nabla U$$

$$|\vec{L}| = m \vec{v} \times \vec{r} \\ = m r^2 \dot{\varphi}$$

$$\omega = v/r$$

$$\varphi = \omega t$$

$$= m r^2 \omega$$

$$= \omega$$

$$m r^2 \omega = A$$

$$m a^2 e$$

$$Z = r e^{i\omega t}$$

$$\frac{dZ}{dt} = \frac{\partial r}{\partial t} \frac{\partial Z}{\partial r}$$

$$\left(\frac{\partial Z}{\partial t}\right)^2 = \left(\frac{\partial r}{\partial t}\right)^2 \left(\frac{\partial Z}{\partial r}\right)^2 \frac{1}{Z}$$

$$= \frac{1}{r} \left(\frac{\partial r}{\partial t}\right)^2 e^{i\omega t}$$

$$\vec{r} = r \hat{e}_r$$

